Staying Power: The Individual Effects of the H-1B Visa Lottery*

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Abstract

TODO

1 Introduction

Skilled immigration sits at the center of several debates in labor economics. One set of questions concerns the effects of high-skill immigration on firms, innovation, and the employment of native workers. Another concerns the private consequences of immigration policy for the migrants whose careers are directly shaped by visa rules. The H-1B program is especially important in both debates because it is the main capped channel through which U.S. employers hire foreign workers in specialty occupations. Yet while a large literature studies how changes in H-1B access affect firms and local labor markets, much less is known about what happens to the individual worker who wins or loses the H-1B lottery. That missing worker-level margin matters for both policy and welfare. If losing the lottery typically forces workers to leave the sponsoring employer or to leave the United States altogether, then the

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lottery creates large private disruptions even if firm-level responses are muted. If instead many losers remain through alternative channels, then the H-1B cap may operate less as a sharp entry barrier and more as a source of delay and uncertainty.

Estimating those individual effects is difficult for two reasons. First, visa receipt is usually endogenous to labor-market outcomes. Firms apply for workers they value, and workers who obtain visas differ systematically from those who do not. The H-1B cap lottery offers a natural solution to that problem, but only if one can observe both winners and losers. Second, the administrative data that record H-1B applications are typically deidentified. In the absence of names or stable person identifiers, it is challenging to follow individual applicants into longitudinal labor-market data, especially for lottery losers, who may disappear from standard U.S. administrative sources if they leave the country or never transition onto H-1B status. Existing work has therefore focused largely on firm-level outcomes or on selected employer panels rather than on the individual applicant.

This paper overcomes both obstacles. I combine the universe of recent cap-subject H-1B lottery records obtained through FOIA requests with LinkedIn-based individual work-history data from Revelio Labs. The empirical leverage comes from a change in the institutional environment surrounding the H-1B program. In the older view of the program, many applicants were abroad at the time of filing, so employers only observed workers inside the firm if the petition succeeded. In the modern setting, a large share of applicants are already working in the United States before the lottery, most commonly under F-1 Optional Practical Training (OPT). For this group, the sponsoring firm often employs the worker before it enters the lottery. That fact makes it possible to condition on pre-lottery employment, define a credible candidate pool inside the sponsoring firm, and link deidentified H-1B applicants to individual LinkedIn profiles using probabilistic record linkage.

The first contribution of the paper is methodological. I develop a linkage procedure tailored to deidentified immigration records. The procedure begins with an employer crosswalk between the administrative H-1B files and Revelio firms, combining exact and fuzzy company-name linkage with LLM-assisted review of ambiguous cases. On the Revelio side, I then impute the sparse fields needed for individual linkage—country of origin, year of birth, sex, and advanced-degree status—using education histories, firm histories, and name-based

models. The resulting applicant-candidate scores are converted into within-applicant match weights, which I carry forward into the outcome analysis instead of forcing a single deterministic match. This feature is useful in precisely the setting studied here, where no single identifier is decisive and where researchers would like to preserve uncertainty rather than hide it.

The second contribution is substantive. To the best of my knowledge, this is the first paper to estimate the causal effect of winning the H-1B lottery on the individual worker using application-level administrative data on both winners and losers. My empirical design compares applicants who were selected and not selected in the lottery, conditioning on advanced-degree-exempt status because U.S. master's and doctoral degree holders face a different lottery path. In the linked sample, I collapse candidate-level outcomes to the applicant level using the probabilistic match weights and estimate specifications with and without firm-by-lottery-year fixed effects. Identification relies on the standard lottery assumption that, conditional on advanced-degree status, selection is independent of applicant potential outcomes.

The current estimates point to a consistent but modest effect of winning. Lottery winners are more likely to remain with the sponsoring employer and more likely to be observed working in the United States in the years after the lottery. In the current draft, the effect on same-firm retention two years after the lottery is about 0.9 percentage points, and the effect on remaining in the United States is about 2.8 percentage points. These effects are economically meaningful but small relative to baseline attachment. The magnitudes appear to reflect two forces. First, probabilistic linkage inevitably introduces measurement error, which attenuates treatment effects toward zero. Second, the small estimates seem to be substantive rather than purely mechanical: even in high-quality matched subsamples, many lottery losers remain at the firm or in the United States through other channels, including STEM OPT extensions, subsequent lottery attempts, cap-gap protection, or alternative visa paths. The effects are correspondingly larger for workers who appear to be closer to the end of their OPT runway.

These findings speak to several literatures. A long tradition studies the aggregate and firm-level consequences of skilled immigration policy, including innovation, productivity, and

employment effects of changes in H-1B access (Bound et al., 2017; Kerr and Lincoln, 2010; Doran et al., 2022; Mayda et al., 2020; Mahajan et al., 2024). A related line of work emphasizes how immigration rules can shape firms' outside options and the global geography of work, for example by inducing offshoring or affecting destination choices outside the United States (Glennon, 2024; Khanna and Morales, 2017; Brinatti and Guo, 2023). Another literature highlights that temporary work visas may generate employer-specific rents by restricting worker mobility and tying legal status to the sponsoring employer (Kim and Pei, 2022). This paper complements those studies by shifting the unit of analysis from the firm to the worker. It asks not how an additional H-1B affects the sponsoring firm, but how winning the lottery changes the path of the individual applicant.

The results also suggest a different interpretation of the modern H-1B program for a large and growing subset of applicants. Among U.S.-trained workers already employed under OPT, the H-1B often functions less as a mechanism for initial entry and more as a mechanism for retention and longer-horizon attachment. That distinction matters for how one interprets both efficiency and distributional concerns. If many applicants are already present and productive in the United States before the lottery, then the main effect of lottery loss may be to weaken firm attachment, reduce employment stability, or accelerate return migration only at the margin. At the same time, the modest average effects should not be read as evidence that the lottery is unimportant. A policy that creates substantial uncertainty, repeated reapplication costs, and employer dependence can have large welfare consequences even when average reduced-form employment responses are numerically small.

The remainder of the paper proceeds as follows. Section **??** describes the institutional setting, with particular attention to the H-1B lottery and the role of OPT. Section 3 introduces the administrative H-1B records and the Revelio outcome data. Section 4 describes the probabilistic linkage procedure and the construction of the linked applicant panel. Section 5 lays out the empirical design and organizes the results. Section 6 concludes.

2 Background and Context: The H-1B Visa Program

Since its inception in [year], the H-1B visa has remained the primary pathway for temporary skilled workers to immigrate to the US. While the number of H-1B visas issued annually has remained capped at 85,000 for the past x years, the number of applications for H-1B visas (plotted in Figure x) has grown substantially over time. In years where demand exceeds supply (as has been the case in every year since x), USCIS conducts a lottery to allocate the limited number of visas.

Every year since x, the number of applications has exceeded the cap, prompting USCIS to conduct a lottery to allocate the limited number of visas, which is the source of random variation I use in this paper.

2.1 The H-1B Visa

The H-1B visa is an employer-sponsored temporary work visa for workers in "specialty occupations", created in x as a way for firms to fill skill shortages in the US labor market.

There are three key institutional details of the H-1B visa that are important for this paper. First, the H-1B is employer-sponsored, meaning that employers must apply for the visa on behalf of the worker, and the worker's legal status is tied to the sponsoring employer. While the visa is portable between employers,... fill in. Workers cannot self-sponsor.

Second, the H-1B is temporary, with a maximum duration in most cases of six years (three years of initial status plus one three-year extension). Finally, the H-1B is dual-intent, meaning that while it does not lead directly to permanent residency, it does not preclude workers from applying for permanent residency while on H-1B status. Many H-1B workers apply for employer-sponsored green cards under their H-1B employer, and while H-1B visas are portable across employers, pending green card applications are not, creating an additional incentive for workers to remain with their sponsoring employer until they receive their green card.

For employers to sponsor an H-1B worker, they must first file a Labor Condition Application (LCA) with the Department of Labor, attesting that they will pay the worker the prevailing wage for the occupation and location, and that the employment of the H-1B

worker will not adversely affect the working conditions of similarly employed US workers. After receiving LCA certification, employers can then file Form I-129 with USCIS to petition for the H-1B visa on behalf of the worker. Put this in 'petition' language.

2.2 The H-1B Lottery

Since x, the number of H-1B visas that can be allocated in any given year has remained capped at 85,000. This is partitioned into 65,000 visas for all applicants ('regular' cap) and an additional 20,000 visas reserved for applicants with an "Advanced Degree Exemption" ('ADE' cap), i.e. applicants with an advanced degree (Master's or higher) from a U.S. institution. Certain types of institutions fill are exempt from the cap.

Historically, allocation of the cap-subject H-1B visas has been on a first-come, first-served basis: applications are accepted and processed on a rolling basis starting on April 1 of each year, and once the cap is reached, USCIS stops accepting applications. In every year since 2013, however, the cap has been reached on the first day of the application period, prompting USCIS to conduct a lottery to allocate the limited number of visas. Starting in FY 2020, USCIS implemented an electronic registration system for the H-1B lottery. Prior to this system, firms were required to submit the full petition at the time of application, and USCIS would conduct the lottery among all submitted petitions *without retaining any information on petitioners not selected in the lottery.*

Under the new system, firms first submit a short electronic registration during a registration window in March. USCIS then conducts the random lottery separately by ADE status and notifies selected registrants (henceforth lottery 'winners').[1] Lottery winners are then given a window of 90 days to submit a fully completed petition for an H-1B application, including the labor condition application and supporting documentation. If that petition is approved, the earliest start date for cap-subject employment is October 1 of the relevant fiscal year.

This new registration system substantially reduced the cost of application for employers

[1] The ADE lottery is conducted first among ADE-eligible applications. The regular lottery is subsequently conducted with all non-ADE applications **and** ADE applications not selected in the ADE lottery. ADE applications can therefore be selected through either of the ADE or non-ADE lotteries.

(who no longer need to submit a full petition for every applicant) and created an electronic record of lottery applicants, including lottery losers, the crucial source of data for this paper. In reducing the cost of application, however, the new system also created scope for multiple registrations on behalf of the same beneficiary. During the years in my sample, the random selection operated at the registration level rather than at the beneficiary level, meaning that a worker with multiple registrations could receive more chances in the lottery. USCIS later adopted a beneficiary-centric selection process in response to these integrity concerns, beginning after the main years I study (U.S. Citizenship and Immigration Services, 2024a, 2025a). This matters empirically because duplicate-heavy employers and outsourcing-style applications can distort both the interpretation of the lottery and the quality of individual linkage. I therefore describe the multiple-registration issue explicitly in the data section and treat it as both an institutional feature of the sample period and a source of robustness checks in the linked analysis.

2.3 Optional Practical Training (OPT) and the Pre-H-1B Worker

The central institutional fact behind the linkage strategy is the importance of Optional Practical Training (OPT). Students on F-1 visas can typically work for up to 12 months after completing their studies through post-completion OPT, and eligible STEM graduates can apply for a 24-month extension, yielding up to 36 months of work authorization in total (U.S. Immigration and Customs Enforcement, 2024; U.S. Citizenship and Immigration Services, 2024b). For many international graduates of U.S. universities, the typical sequence is therefore: graduate, begin working immediately for a U.S. employer under OPT, and then enter the H-1B lottery one or more times while already employed.

This has three consequences for the analysis. First, many lottery applicants are already attached to the sponsoring employer before the lottery takes place. That creates a feasible linkage design, because the relevant candidate pool in LinkedIn data is not "everyone in the world with similar demographics" but rather "workers already observed at the sponsoring firm around the lottery date." Second, lottery loss need not imply immediate separation from the employer or immediate exit from the United States. A worker who loses can remain under the initial 12-month OPT period, under the STEM extension, through cap-gap protection

if a later petition is pending, or through another temporary status (U.S. Citizenship and Immigration Services, 2025b). Third, the treatment effect of winning should vary with remaining runway on OPT. The difference between winning and losing is likely to be largest when alternative work authorization is about to expire.

3 Data

3.1 Administrative Data

My data on H-1B applications was obtained from U.S. Customs and Immigration Services (USCIS) via a Freedom of Information Act (FOIA) request from Bloomberg... [source]. It is a complete record of all cap-subject lottery applications in FY 2020-2023, including the applicant's country of origin, year of birth, gender, sponsoring employer, and lottery outcome. For lottery winners who subsequently filed a petition for an H-1B visa, I have much more detailed information including their college major, job title, previous visa status, etc. The data do not contain indicators for ADE status for lottery losers.

Each row of the H-1B data represents a single lottery application, which is the unit of treatment assignment in the analysis. Importantly, as discussed in Section **??**, the data include many duplicate applications from different employers on behalf of the same applicant. The data include an indicator for whether an applicant is a 'duplicate' (i.e. whether the applicant was already named on another application in the same lottery year), but do not indicate the original application. Therefore while I can filter on this indicator to obtain a sample of applications that are unique at the applicant x year level, I am not able to fully exclude all applicants that submit multiple applications in the same year (since one of the applications will not be marked as a duplicate or indicated in any way). To address this as best as possible, I use the duplicate indicator to identify and exclude employer-year cells with *any* duplicate applications.

3.2 Revelio Data

My data on outcomes... comes from Revelio Labs. Scraped LinkedIn profiles... Position data on individuals' complete work histories (as reported on LinkedIn), including employer, start and end dates, location, job title.

Education data on individuals' complete education histories (as reported on LinkedIn), including institution name, program start and end dates, degree, and field.

The outcome data come from Revelio Labs' large snapshot of public LinkedIn-based profiles. In the current project files, Revelio contains worker-level profiles with names, profile-update dates, education histories, and detailed position histories, along with company identifiers (`rcid`), titles, start and end dates, locations, and imputed compensation for many positions. The raw scale is large enough that the data behave like a near-universe of digitally visible professional workers, which is particularly useful in a study of the skilled labor market.

The main strength of Revelio for this project is that it observes both the pre-lottery and post-lottery histories of workers who are likely to be in the H-1B applicant pool. The pre-period is crucial: it allows me to identify which users were already employed at the petitioning firm before the lottery and to restrict attention to those applicants for whom a LinkedIn-based match is feasible. The post-period is equally important because it allows me to track whether a worker remains at the sponsoring firm, switches firms, appears to leave the United States, or returns to education.

At the same time, Revelio is not a frictionless measure of labor-market outcomes. Profile coverage is selective, profile updates are not continuous, location fields are noisy, and some workers disappear from the platform for reasons unrelated to migration or employment. Those issues motivate two choices in the paper. First, the administrative records determine treatment status and the basic analysis sample; Revelio is used primarily for linkage and outcomes rather than for treatment assignment. Second, I treat probabilistic linkage uncertainty as an object to be propagated forward into the regression analysis rather than as a nuisance to be hidden.

3.3 Analysis sample

The linked analysis is necessarily a subset of the raw H-1B applicant pool. The main sample focuses on applicants who plausibly were already employed at the sponsoring firm before the lottery. This restriction is both substantively natural and empirically valuable. It targets the margin for which winner-loser comparisons are most credible in LinkedIn data and excludes cases where the worker was likely abroad and therefore essentially unobservable absent a successful H-1B transition.

In practice, the linked sample is shaped by three filters. First, the employer must be linkable from the FOIA files into Revelio. Second, there must exist a plausible pool of candidate workers at the linked employer around the lottery date. Third, the applicant-candidate match must pass the probabilistic-linkage filters described in Section 4. Depending on the specification, I further restrict the sample by candidate multiplicity, match quality, or employer-level duplicate-registration patterns. The current linked sample is on the order of 100,000 applicants, but the final paper should report exact counts for the baseline, strict, and alternative samples in a transparent sample-construction table *[insert final Ns by sample variant]*.

3.4 Outcomes

The main outcomes are drawn from Revelio position and education histories. The paper's central retention outcome is an indicator for whether the linked worker is still employed by the sponsoring firm one, two, or three years after the lottery. The corresponding migration outcome is an indicator for whether the worker is observed in the United States over the same horizon. Additional outcomes include switching firms, promotion-like within-firm job changes, new education enrollment, and compensation measures based on Revelio's imputed salary fields.

Two points are useful to emphasize in the final draft. First, the outcome definitions are designed to be conservative with respect to noisy profile timing. For example, same-firm retention is defined at the employer level rather than requiring a specific title match. Second, the "in the U.S." outcome is an observed-location measure rather than a direct legal-status

measure. It should therefore be presented as evidence on observed geographic attachment, not as a literal measure of immigration status. Appendix 7 describes the underlying construction in more detail.

4 Probabilistic Linkage

This section describes the paper's main methodological contribution: a probabilistic linkage between deidentified H-1B administrative records and LinkedIn-based worker histories. The linkage proceeds in three stages. I first build an employer crosswalk between the FOIA files and Revelio firms. I then impute the sparse fields needed for person-level matching on the Revelio side. Finally, within each employer-year cell, I score candidate workers and carry forward the resulting match weights into the outcome analysis.

4.1 Step 1: Employer linkage

The first problem is to align employers in the FOIA data with employer identifiers in Revelio. The code base constructs a firm-year crosswalk using cleaned employer names, employer tax identifiers where available, lottery year, and a union of previously validated matches. Exact or high-confidence matches are kept directly. Ambiguous firm-name cases are first narrowed using fuzzy string comparison and then reviewed using an LLM-assisted workflow that classifies whether a candidate Revelio firm is a valid crosswalk for the FOIA employer. The LLM is therefore used only at the firm-linkage stage, where the task is to evaluate company identities, rebrands, subsidiaries, or spelling variants; it is not used to identify individual workers.

Allowing the employer crosswalk to be many-to-many at the firm-year level is important. A single employer can map to multiple Revelio company identifiers because of rebranding, acquisitions, legal-entity fragmentation, or data-vendor conventions. Conversely, forcing a one-to-one firm mapping too early can create false non-matches that cascade into the person-level linkage. The current pipeline therefore carries forward all valid firm-year crosswalks and later uses additional evidence from workers' position histories to refine match quality.

4.2 Step 2: Constructing Revelio-side linkage variables

Because the administrative H-1B records are deidentified, the person-level match cannot use names directly. Instead, I exploit a set of sparse fields that appear in both datasets either directly or by imputation.

First, I estimate year of birth in Revelio using education timing. When a profile reports a high school degree, the implied birth year is derived from the observed high school end date. Otherwise, the estimate comes from the earliest postsecondary start date or end date, with adjustments for the degree type. Second, I construct a probabilistic sex measure from names, averaging across alternative name-based models to reduce the chance that any one model performs poorly for particular naming conventions.

Third, I infer country of origin using a combination of information from education histories, names, and career histories. Institution-country matches use the locations of schools and universities, while name-based country and subregion predictions come from two different models (a country classifier and a broader subregion classifier). Position histories provide an additional consistency check when a worker's pre-lottery job history includes the same origin country observed in the administrative record. The current code combines these ingredients into a country score and a subregion score, while also flagging cases where the name-based predictions are especially uncertain.

Fourth, I use education histories to impute variables that matter later in the empirical design, even if they are not primary linkage fields. These include an indicator for advanced-degree exemption status, the year of the latest U.S. graduation, and field-of-study measures that help identify STEM-oriented profiles. These variables are useful both substantively and operationally: they sharpen the match and help define heterogeneity by likely remaining OPT runway.

Finally, I deduplicate Revelio profiles when multiple profiles appear to correspond to the same underlying person. The current implementation treats profiles as duplicates only when they share a normalized name and an employer identifier and also satisfy an overlap condition based on job dates, education dates, or school names. Within duplicate groups, the profile with the richer position history is retained. This step is important for avoiding

overcounting in large employers with many similar profiles.

4.3 Step 3: Candidate pools and applicant-candidate scoring

For each H-1B applicant, I define the candidate pool as workers who appear in Revelio at the linked employer around the lottery date. This is the key economic restriction that makes the individual-level linkage feasible. In older H-1B settings, employer-based linkage would be biased because many workers were not yet at the firm before the visa decision. In the modern OPT setting, however, pre-lottery employment at the sponsoring firm is common rather than exceptional.

Within this candidate pool, I compute an applicant-candidate score that combines agreement on country of origin, year of birth, and sex, along with measures of firm-link confidence and occupational plausibility. In the current code, the baseline score takes the form

$$S_{ic} = m_{ic}^{\text{firm}} \left[(1 - \omega_{occ}) \left(\omega_{country} s_{ic}^{country} + \omega_{yob} s_{ic}^{yob} + \omega_{sex} s_{ic}^{sex} \right) + \omega_{occ} s_{ic}^{occ} \right], \tag{1}$$

where i indexes applicants and c indexes candidate Revelio workers. The current baseline weights in the code are $\omega_{country} = 0.70$, $\omega_{yob} = 0.20$, and $\omega_{sex} = 0.10$, with the occupation component used only in some variants. The firm-quality multiplier m_{ic}^{firm} equals one when the candidate's position history directly supports the employer crosswalk and otherwise scales with the normalized firm-match score from the employer-linkage stage. Appendix Table 1 reports the current defaults used by the working code.

The scoring stage also imposes several plausibility filters. Candidate workers must have start and end dates that make it plausible they were employed at the sponsoring firm around the lottery date; age implied by the estimated birth year must be consistent with the administrative year-of-birth field; and, where observed, education and occupational histories must look broadly consistent with H-1B-eligible work. When no candidate achieves a sufficiently strong country-level match, the algorithm allows a controlled fallback to subregion-level evidence, but only when the overall score is still credible.

4.4 From scores to probabilistic match weights

A central choice in the paper is not to force a deterministic one-to-one person match. Instead, I convert the applicant-candidate scores into within-applicant weights,

$$w_{ic} = \frac{S_{ic}}{\sum_{c' \in C_i} S_{ic'}}, \tag{2}$$

where C_i is the set of retained candidates for applicant i. These weights sum to one within applicant and capture the relative likelihood that each candidate is the correct match.

This choice matters for the econometrics. If the linkage were forced to select a single candidate in every case, mistakes would enter the outcome data in a difficult-to-model way and would likely generate both attenuation and classical misclassification. By preserving multiple candidates, the analysis can instead work with applicant-level weighted outcomes. In the current regression code, each outcome is first aggregated to the applicant level using the w_{ic} and then regressed on lottery-selection indicators. The resulting estimand is naturally conservative: if linkage is noisy, treatment effects are attenuated toward zero rather than driven by arbitrary hard assignments.

4.5 Quality controls, strict samples, and validation

Because probabilistic linkage can still fail in systematic ways, the analysis uses several quality controls. First, I flag applications for which the top candidate is only weakly separated from the second-best candidate. Second, I estimate results in samples that limit candidate multiplicity. Third, I define a strict high-precision sample that requires a dominant top weight, a high absolute match score, strong firm-link quality, strong country evidence, and a nonmissing estimated birth year. These restrictions are designed to reduce false positives at the cost of recall.

The working code currently produces a family of analysis variants—including a baseline sample, samples that cap the number of retained candidates per applicant, and a strict sample—so that the paper can transparently show how sensitive the estimates are to linkage quality. In the final draft, I recommend emphasizing two comparisons. The first is between the baseline weighted results and the strict high-precision sample, which speaks directly to

attenuation from match error. The second is between the linked sample and employer-year cells with especially clean applicant coverage, which addresses concern that selection into linkage may vary with treatment.

A fuller account of the linkage pipeline, including the exact thresholds used in the current code and the construction of the outcome variables, is provided in Appendix 7.

5 Results

This section is intentionally structured as a template for the full results discussion. The empirical design and the likely interpretation of the estimates are laid out below, but the detailed tables and figures can be inserted once the preferred sample and exact specifications are finalized.

5.1 Empirical specification

The main specification compares lottery-selected and non-selected applicants in the linked sample, conditioning on advanced-degree-exempt status because ADE applicants face a different lottery assignment process. Let Win_i indicate lottery selection for applicant i, and let ADE_i indicate eligibility for the advanced-degree exemption. For an applicant-level outcome y_i, the baseline estimating equation is

$$y_i = \beta_0 + \beta_1 Win_i + \beta_2 ADE_i + \beta_3 (Win_i \times ADE_i) + \alpha_{f(i),s(i)} + \varepsilon_i, \tag{3}$$

where $\alpha_{f(i),s(i)}$ denotes firm-by-lottery-year fixed effects in the preferred specification. In the current code, outcomes are first aggregated to the applicant level using the probabilistic match weights described in Section 4, and standard errors are clustered at the firm level.

The parameter β_1 captures the effect of winning for regular-cap applicants, while $\beta_1 + \beta_3$ captures the effect for ADE applicants. In practice, I recommend presenting the regular-cap estimate, the ADE estimate, and a pooled marginal effect if the two are similar enough to justify a summary statistic. A short subsection or appendix table should also document balance on pre-lottery observables within ADE groups, since that is the most transparent

way to reassure readers that the lottery randomization is behaving as expected in the linked sample.

5.2 Main retention effects

The first set of outcomes should focus on same-firm retention one, two, and three years after the lottery. These are the cleanest outcomes conceptually because they map directly onto the employer-tie embedded in the H-1B program. The natural table structure is a column for each horizon and a row for the regular-cap and ADE effects. A companion event-study-style figure could also be useful if the relative-year outcome construction is expanded beyond the first three post-lottery years.

The current draft's headline result is that lottery winners are modestly more likely to remain at the sponsoring employer. A concise interpretation paragraph could read as follows once the exact table is inserted: winning the H-1B lottery increases short- to medium-run attachment to the sponsoring firm, but the effect is small relative to baseline retention. That pattern is consistent with the idea that H-1B status increases employer-specific attachment, yet many non-selected workers also remain at the firm for at least some time because they continue under OPT or another temporary arrangement. *[Insert preferred retention table and exact magnitudes.]*

5.3 Remaining in the United States

The next set of results should turn to whether the worker remains observed in the United States. This outcome is particularly important because many policy arguments implicitly treat H-1B rationing as a yes-or-no margin for U.S. employment. The current estimates suggest that winners are more likely to remain in the United States two years after the lottery, but again the magnitudes are modest.

A useful way to write this section is to emphasize both the sign and the size of the estimates. The sign is informative: winning stabilizes U.S. attachment. The size is equally informative: the fact that the effect is not larger suggests that a substantial share of lottery losers continue to work in the United States through other channels. That interpretation

should be tied back to OPT, repeat lottery participation, and the possibility of alternative legal statuses. *[Insert U.S.-presence estimates and supporting figure or table.]*

5.4 Career adjustment: firm switching, promotion, and education

The third block of results can examine mechanisms and adjustment margins. The most natural outcomes are switching employers, within-firm promotion-like job changes, enrollment in additional education, and compensation measures. These outcomes help distinguish between several interpretations of the main findings.

For example, if losers are less likely to stay at the original firm but are not much less likely to remain in the United States, then adjustment may occur largely through employer switching. If losers are more likely to return to school, that would support the view that additional education is used as a way to preserve legal status or buy time for a later H-1B attempt. If winners exhibit lower rates of new education but similar wage paths, that would suggest the main role of the H-1B is legal stability rather than an immediate productivity or pay shock. *[Insert mechanism table once preferred outcomes are finalized.]*

5.5 Heterogeneity by remaining OPT runway

A particularly important heterogeneity dimension is the amount of time the worker appears to have remaining before OPT exhaustion. The code already constructs a measure of time since graduation, which is a natural proxy for remaining OPT runway. The prediction is straightforward: effects of winning should be larger for workers who are closer to the end of their allowable OPT period.

I recommend presenting this in two ways. First, split the sample into broad groups, such as workers with short versus long remaining runway. Second, report an interaction specification in which lottery selection is interacted with an indicator for being near the end of OPT eligibility. This section is likely to be one of the most persuasive pieces of evidence that the modest average effects are not purely driven by noise. If the treatment effect grows when alternative authorization is scarce, that is exactly what the institutional context would predict. *[Insert heterogeneity table/figure.]*

5.6 Robustness to linkage quality and sample selection

Given the novelty of the linkage, the robustness section should be prominent rather than perfunctory. At minimum, the final paper should compare estimates across: (i) the baseline weighted sample, (ii) high-precision or strict matches, (iii) samples that cap candidate multiplicity, and (iv) employer-year cells with especially complete applicant coverage. A compact summary table with the same headline outcome repeated across linkage variants is often more convincing than a long sequence of appendix tables.

This is also the right place to address the concern that the linked sample may not be representative of all H-1B applicants. The key distinction is between internal and external validity. Internal validity requires that, conditional on ADE status, winning does not differentially affect inclusion in the matched sample. External validity concerns which subset of applicants the paper speaks to most directly—namely, workers who were already employed and sufficiently visible in LinkedIn data before the lottery. The discussion should be explicit that the paper's most credible interpretation is for that pre-H-1B worker margin rather than for all potential H-1B beneficiaries worldwide.

5.7 Interpreting small magnitudes

The conclusion of the results section should directly confront the central substantive fact of the paper: the effects are positive and statistically meaningful, but small. I would frame the interpretation around two non-exclusive explanations.

First, probabilistic linkage error attenuates reduced-form treatment effects. Even if the treatment effect on the true matched worker is moderate, mixing in incorrect candidates will mechanically compress the estimate toward zero. Second, and more interestingly, the small estimates appear to be economically real. Among U.S.-trained applicants already employed under OPT, the H-1B lottery is often a retention lottery rather than an entry lottery. Losing the lottery increases separation risk and the risk of leaving the United States, but many workers still find a way to remain. That interpretation is consistent with the heterogeneity by OPT runway and with robustness checks on higher-quality matches.

A short final subsection or paragraph can then bridge to the conclusion: for this sub-

group of applicants, the H-1B matters, but its main effects are on stability and persistence rather than on an immediate binary transition from employed in the United States to absent from the U.S. labor market.

6 Conclusion

This paper asks a simple but previously difficult question: what does winning the H-1B lottery do to the individual worker? I answer that question by combining recent deidentified H-1B administrative records with LinkedIn-based worker histories from Revelio Labs and by developing a probabilistic linkage strategy that can connect the two without direct personal identifiers. The approach is designed around the institutional reality that many recent H-1B applicants are already working for the sponsoring employer before the lottery, often under OPT. That fact makes the worker-level question empirically tractable.

The paper makes two main contributions. First, it introduces a linkage framework for deidentified immigration records that should be useful beyond this specific application. The contribution is not only that a link can be made, but that the uncertainty in the link can be carried transparently into the analysis through match weights and high-precision subsamples. Second, it provides the first worker-level causal evidence, to my knowledge, on the consequences of winning the H-1B lottery. The current results indicate that lottery winners are more likely to remain with the sponsoring employer and more likely to remain in the United States, but the magnitudes are small.

Those small effects are themselves informative. They suggest that, for a large subset of recent applicants, the H-1B cap is not simply a gate that cleanly separates entry from non-entry. Instead, it interacts with a broader system of temporary work authorization in which many lottery losers continue to work, reapply, extend OPT, or transition through other channels. For the modern pre-H-1B worker, winning appears to increase labor-market stability more than it changes the extensive margin of being present in the United States at all.

This interpretation has two policy implications. The first is that reforms to the H-1B system should be evaluated not only in terms of how many workers they admit, but also in

terms of how they change employer dependence, repeated application costs, and uncertainty for workers already trained and employed in the United States. The second is that the individual-level incidence of H-1B rationing may differ from the incidence suggested by firm-level studies: the main worker-level effects may operate through retention and mobility rather than through abrupt displacement.

There are several natural extensions. One is to study wage and career progression in more detail once the preferred compensation measures are finalized. Another is to connect the linked applicant panel to productivity outcomes, such as GitHub activity or entrepreneurial transitions. A third is to study return migration and third-country reallocation more directly. More broadly, the project shows that the deidentified nature of administrative immigration data, while a serious obstacle, need not prevent credible worker-level analysis when combined with modern linkage methods and a careful understanding of institutional context.

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7 Detailed Linkage Appendix

This appendix provides a more detailed description of the linkage pipeline used to connect deidentified H-1B administrative records to Revelio worker histories. The discussion is written to be close to the current code base, so some thresholds and sample labels should be updated if the final production code changes.

7.1 A. Employer crosswalk construction

The employer-linkage problem is solved before any person-level matching takes place. The current pipeline builds a firm-year crosswalk from the FOIA employer identifier (`foia_firm_uid`) to Revelio company identifiers (`rcid`) using three sources of evidence:

1. **Exact or legacy validated matches.** Previously validated FOIA-to-Revelio matches are retained directly and merged into the current employer-year panel.

2. **Fuzzy candidate generation.** Cleaned employer names are compared across the FOIA and Revelio data using string similarity, tax identifiers when available, and lottery year.

3. **LLM-assisted review of ambiguous firm matches.** Ambiguous cases are reviewed using an LLM workflow that labels whether the candidate Revelio firm is a valid match. This step is restricted to firm identity resolution; it does not operate on worker identities.

The resulting crosswalk allows multiple valid `rcid` values for a given employer-year. In the current implementation, the code keeps one best row per employer-year-`rcid` combination while also ranking `rcid`s by match quality. This flexibility is important because vendor company identifiers may split or consolidate the same economic employer across years.

7.2 B. Defining the linked-worker universe in Revelio

The linkage deliberately focuses on workers who are likely to be in-scope for the H-1B lottery. The current Revelio cleaning script begins by restricting attention to users attached to

employers in the in-sample H-1B employer-year panel and to U.S.-based positions beginning after a conservative calendar cutoff. It then cleans position histories, imputes end dates for some open positions, and removes duplicate profiles when they appear to represent the same person.

The duplicate-profile step is conservative. Profiles are considered duplicates only when they share a normalized name and employer identifier and also satisfy at least one overlap condition: overlapping position dates, overlapping education dates, or common school names. Within confirmed duplicate groups, the profile with the richer position history is kept. This design tries to avoid collapsing genuinely different workers who happen to have the same name and work for the same employer at different times.

7.3 C. Imputing person-level linkage fields in Revelio

The deidentified administrative files do not contain names, so the person-level match relies on sparse fields that are available or inferable on both sides. The current Revelio cleaning pipeline constructs the following variables:

Estimated year of birth. Estimated year of birth is derived from education timing. If a high school record is observed, the code backs out birth year from the high school end date. Otherwise it uses the earliest postsecondary start or end date, adjusting the assumed age at completion by degree type.

Sex. The code computes a female-probability score from name-based models and compares that probability with the binary sex field in the administrative data. The resulting sex-match score is highest when the two sources agree strongly and is lower in ambiguous cases.

Country and subregion. Country evidence comes from several sources: school-country matches, full-name and first/last-name country predictions, a subregion classifier, and position-history information. These inputs are combined into a country score and a subregion score. The code also flags cases with especially uncertain name-based country predictions, which later affects candidate ranking.

Advanced-degree status and graduation timing. Using U.S. education records, the code constructs an indicator for whether the user appears to hold a U.S. master's, MBA, or doctoral degree, along with an imputed advanced-degree year and latest U.S. graduation year. These variables are later used for both matching and heterogeneity analysis.

Occupational plausibility. Position titles are cleaned and crosswalked to an H-1B-relevant occupation ranking. The pipeline can use this information as an optional occupation score in the candidate ranking stage.

7.4 D. Candidate generation within employer-year

For each H-1B applicant, the candidate pool is defined inside the linked employer-year cell. Candidate workers must satisfy timing conditions that make it plausible they were already employed at the sponsoring firm around the lottery date. In the current code, workers are retained only if their start and end dates place them near the March lottery reference date, if the implied age from the estimated birth year is compatible with the administrative birth year, and if the profile looks broadly consistent with STEM-oriented or H-1B-like employment.

This employer-based restriction is central to the design. It greatly reduces false matches relative to a global search, and it is substantively justified because many modern H-1B applicants are already employed by the sponsoring firm under OPT before the lottery takes place.

7.5 E. Scoring, weighting, and ambiguity flags

After the candidate pool is defined, the code assigns each applicant-candidate pair a total score. The score uses a weighted combination of agreement on country, year of birth, and sex, optionally augmented by occupation similarity and then multiplied by a firm-quality term. Candidate weights are normalized within applicant and later used to aggregate outcomes to the applicant level.

The code also records several diagnostics that are useful for robustness exercises:

- the number of retained candidates per applicant,
- the top candidate's normalized weight,
- the gap between the top and second-best candidate,
- the underlying country, firm, and total scores, and
- whether the application is flagged as ambiguous because the top two candidates are too close.

These diagnostics motivate the strict sample and the multiplicity-based robustness checks.

Table 1: Illustrative linkage parameters in the current code base

Parameter	Current value	Role in linkage
Country weight $\omega_{country}$	0.70	Baseline weight on country-of-origin agreement in total score.
Year-of-birth weight ω_{yob}	0.20	Weight on compatibility between administrative and imputed birth year.
Sex weight ω_{sex}	0.10	Weight on agreement between administrative sex and name-based sex probability.
Occupation weight ω_{occ}	0.10 in some variants	Optional weight on occupation ranking based on Revelio position titles.
Ambiguity weight-gap cutoff	0.03	Flags applications for which the top candidate is not well separated from the runner-up.
Subregion fallback minimum score	0.20	Minimum subregion evidence required when country-level evidence is weak.
No-country minimum total score	0.12	Absolute score floor for cases relying on subregion rather than country evidence.
Strict minimum normalized weight	0.90	Requires the top candidate to account for at least 90% of retained score mass.
Strict minimum total score	0.55	Absolute match-quality floor in the strict high-precision sample.
Strict minimum firm quality	0.80	Requires strong employer-link evidence in the strict sample.
Strict minimum country score	0.50	Requires strong country-level evidence in the strict sample.

7.6 F. Analysis variants generated by the code

The current analysis pipeline produces several parquet variants that correspond to different linkage-quality filters. The baseline sample keeps all retained candidates after the main plausibility filters. Additional variants cap the number of retained candidates per applicant (for example, at 2, 4, or 6), while the strict sample keeps only the top-ranked candidate when it satisfies high-confidence thresholds on weight, total score, employer-link quality, and country evidence. These variants are useful because they separate two questions: whether the sign of the effect is robust to stricter matching, and how much the magnitude grows as false matches are reduced.

7.7 G. Outcome construction from Revelio histories

Once candidate workers are linked, the code constructs relative-year outcomes from Revelio position and education histories. The position-history outcomes are built by comparing each post-lottery year with the reference employer and reference position observed around the lottery date. The education outcomes are built analogously from school histories.

Table 2: Outcome concepts constructed in the current merge pipeline

Outcome	Definition in words
`still_at_firm`t	Indicator that the worker has any active position at the sponsoring employer in year t relative to the lottery.
`change_company`t	Indicator that the worker starts a new position at a different employer after the lottery.
`promoted`t	Indicator that the worker starts a new position at the same employer after the lottery, consistent with a promotion or internal move.
`in_us`t	Indicator that at least one position in year t is located in the United States.
`in_home_country`t	Indicator that at least one position in year t is located in the worker's administrative origin country.
`new_educ`t	Indicator that the worker starts a new education spell after the lottery.
`agg_compensation`t	Year-t compensation measure aggregated across active positions using Revelio's imputed pay and the fraction of the year covered by each job.

7.8 H. Suggested validation exercises for the final draft

I recommend documenting at least three validation exercises in the final paper or appendix.

1. **Holdout-style validation on subsets with extra identifiers.** For some winners, additional education or job fields provide a stronger deterministic check on whether the matched profile is plausible.

2. **Employer-level coverage diagnostics.** Show that results are similar in employer-year cells with especially complete matched coverage and with both winners and losers observed.

3. **Comparison across linkage variants.** Report the same headline outcomes in the baseline, capped-multiplicity, and strict samples to quantify attenuation from linkage

noise.

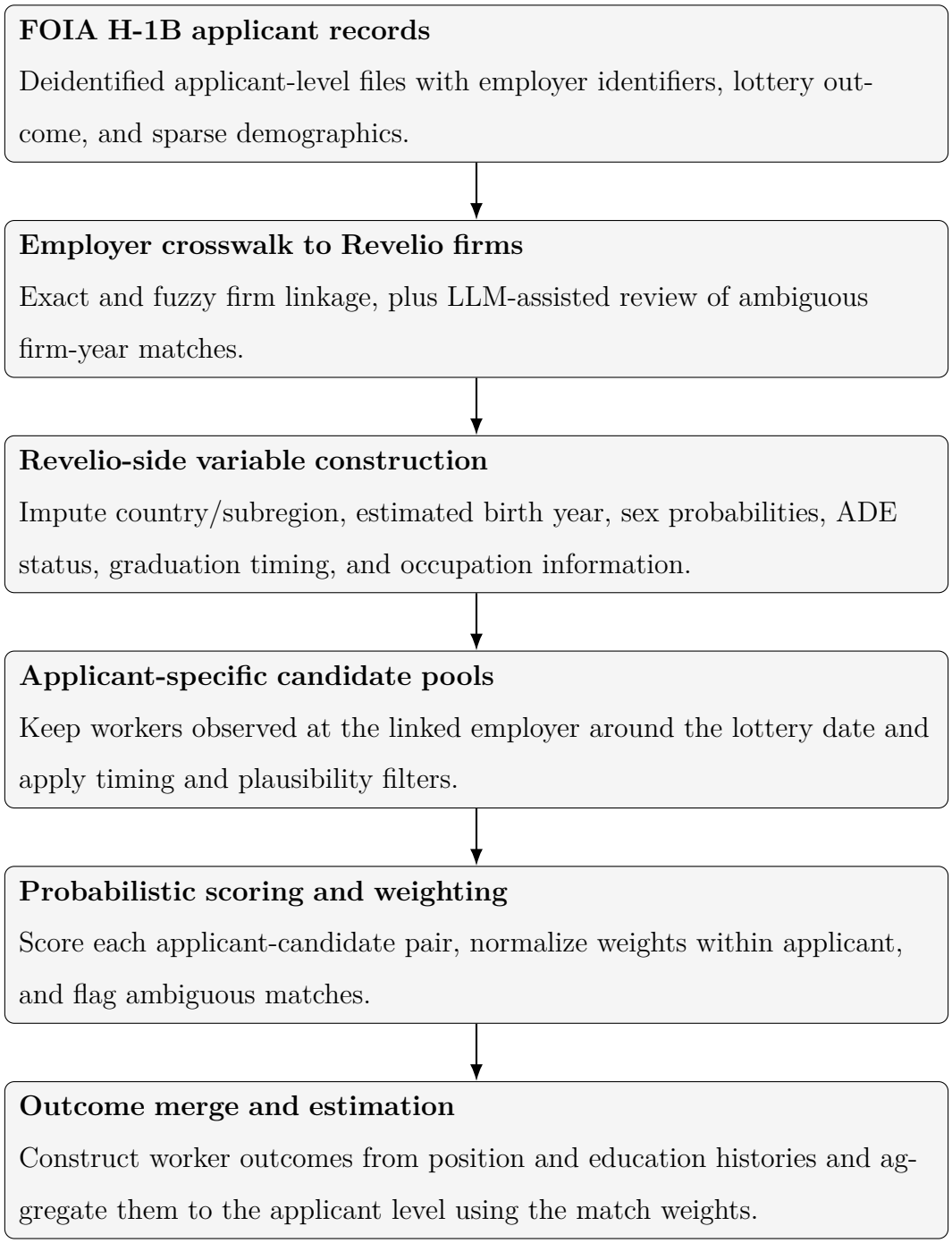
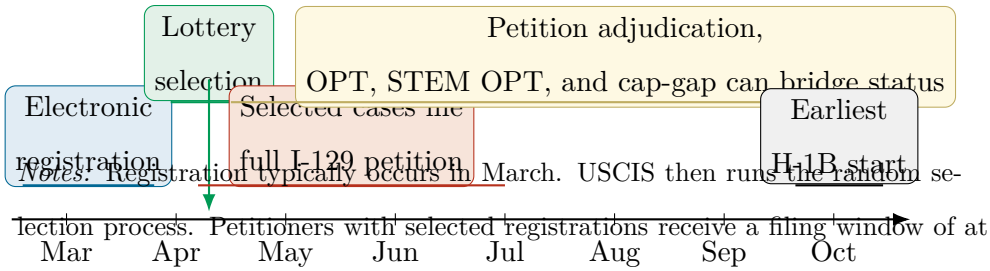


Figure 1: Linkage pipeline from deidentified H-1B records to applicant-level outcomes

8 Detailed Timeline of the Cap-Subject H-1B Process

Figure 2 provides a stylized timeline of the cap-subject H-1B process during the electronic-registration era studied in the paper. The exact calendar dates vary slightly by year, but the sequencing is stable: registration in March, lottery resolution by late March, full petition filing over roughly the next 90 days, and an earliest H-1B start date of October 1. The gap between lottery resolution and October 1 is part of the reason OPT and cap-gap protection matter so much for worker-level outcomes.



Notes: Registration typically occurs in March. USCIS then runs the random selection process. Petitioners with selected registrations receive a filing window of at least 90 days to submit the full petition. For approved cap-subject petitions, the earliest start date is October 1 of the fiscal year. During the years studied here, advanced-degree-exempt applicants faced a distinct lottery path, and many workers were already employed under OPT before the lottery.

Figure 2: Stylized timeline for the cap-subject H-1B process in the electronic-registration era